

Experiment 9: Structured Output Generation – Generate valid Python code and SQL queries through structured prompting techniques

Aim

To generate valid Python code and SQL queries using structured prompting techniques through the OpenAI API.

Python Code:

```
from openai import OpenAI
```

```
client = OpenAI(  
    api_key="sk-proj-Sd2jzBsSZTqHY1pDWZLea6fUdkzFa-  
vQyeUTL2Kdmb1oO8AFZReX80XQrD-  
eDbfpOUyzzDnvnDT3BlbkFJP3FNYcAR4S6TPPtKO-Uw6osECufNvkaZ3kXB-  
WF1xw2XCykfyiZLxwUzLGpzfESUsNJGxcc6YA"  
)
```

```
prompt = """
```

```
Generate the following in a structured format:
```

1. Python Program:

Write a Python program to find the factorial of a number.

2. SQL Query:

Assume a table named STUDENT(ID, NAME, MARKS).

Write an SQL query to display the names of students who scored more than 80 marks.

Display the output with headings:

Python Code:

SQL Query:

```
"""
```

```
response = client.responses.create(  
    model="gpt-5.4-mini",  
    input=prompt  
)
```

```
print(response.output_text)
```

Output

****Python Code:****

```
```python
Program to find the factorial of a number

num = int(input("Enter a number: "))

factorial = 1
for i in range(1, num + 1):
 factorial *= i

print("Factorial =", factorial)
```
```

****SQL Query:****

```
```sql
SELECT NAME
FROM STUDENT
WHERE MARKS > 80;
```

## Result

The structured prompt successfully generated valid Python code and SQL queries in an organized format. The output was accurate, readable, and suitable for programming tasks.